

A Brief Analysis on the World Terrorism – Its Patterns and Trends

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01.

Abstract

Information on more than 180,000 Terrorist Attacks The Global Terrorism Database (GTD) is an open-source database including information on terrorist attacks around the world from 1970 through 2017. The GTD includes systematic data on domestic as well as international terrorist incidents that have occurred during this time and now includes more than 180,000 attacks. The database is maintained by researchers at the National Consortium for the Study of Terrorism and Responses to Terrorism (START), headquartered at the University of Maryland. The analysis commenced with data preprocessing, including handling missing values, removing duplicates, Exploratory data analysis (EDA) techniques were employed to visualize trends and patterns.

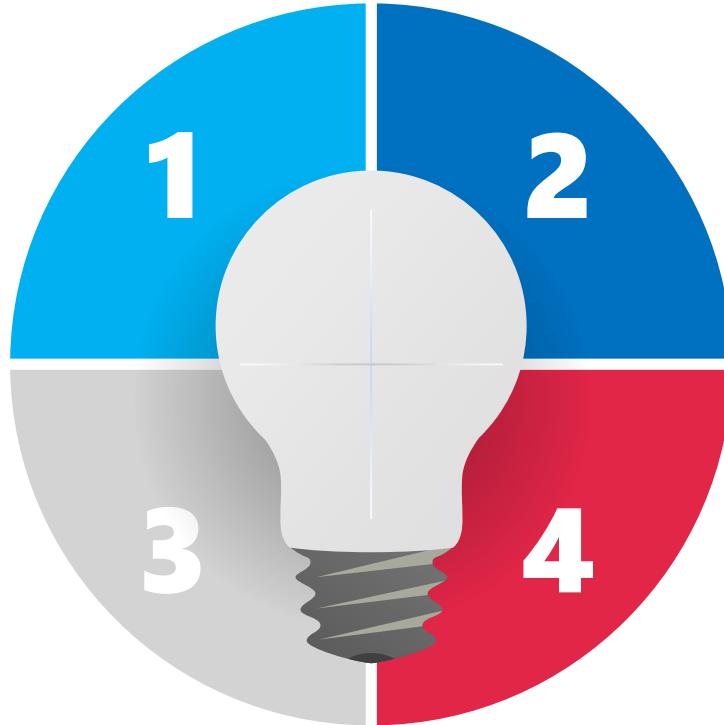
Methodology

Data Loading and Inspection

We'll start by loading the "Global Terrorism Database" and conducting a quick overview of its structure to ensure we're working with the correct data.

Data Visualization

Visualizations play a pivotal role in uncovering insights. We'll create meaningful charts and graphs to visualize trends, patterns, and correlations within the dataset



Data Preprocessing

To prepare the data for analysis, we'll address missing values, outliers, and convert relevant columns into appropriate formats

Conclusion

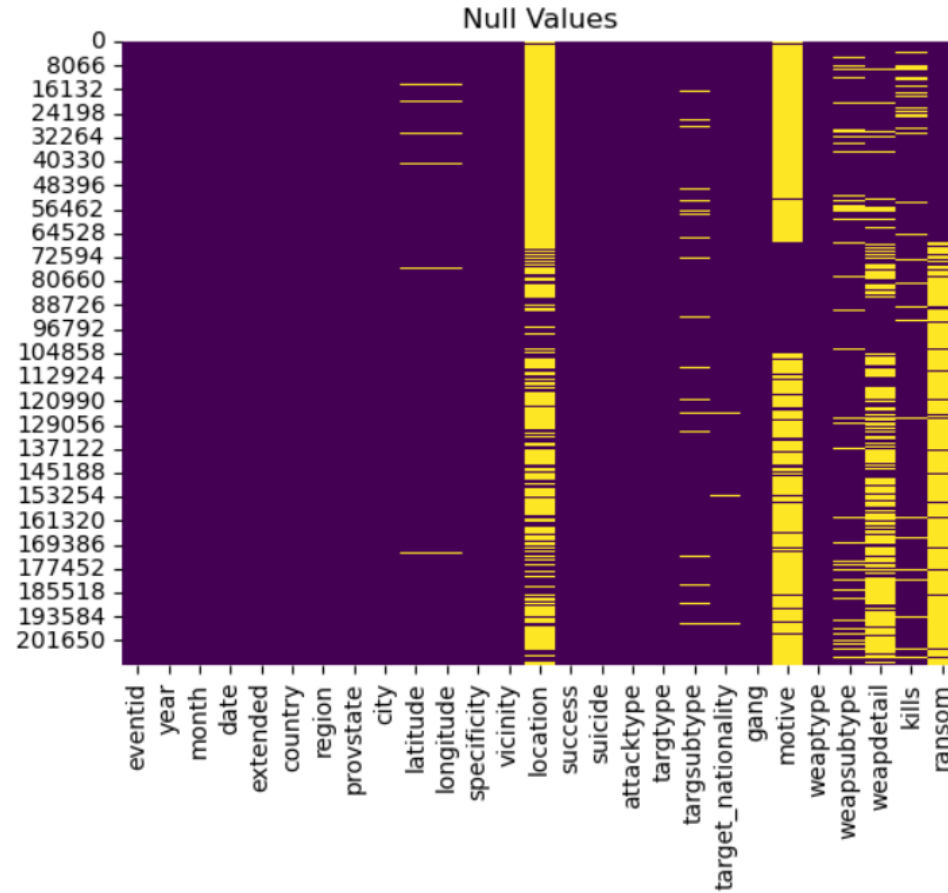
In the end we will conclude the results by identifying trends and patterns.

03.

Dataset

The Global Terrorism Database (GTD) is an open-source database including information on terrorist attacks around the world from 1970 through 2017. The GTD includes systematic data on domestic as well as international terrorist incidents that have occurred during this time and now includes more than 180,000 attacks. The database is maintained by researchers at the National Consortium for the Study of Terrorism and Responses to Terrorism (START), headquartered at the University of Maryland.

Data Cleaning and Preprocessing





Data Cleaning

- NULL Values

Handling NULL values of the following columns

1. Provincial State and City
2. Specificity
3. Target Type
4. Location
5. Weapon type
6. Nationality
7. Motive
8. Weapon details
9. Ransom
10. Kills

04.

Analysis

Trend Analysis: How has the frequency of terrorist attacks changed over the years globally? Further break down this trend by regions to identify specific areas with increasing or decreasing activity

```
In [70]: 1 attacks_by_year = df['year'].value_counts().sort_index()
```

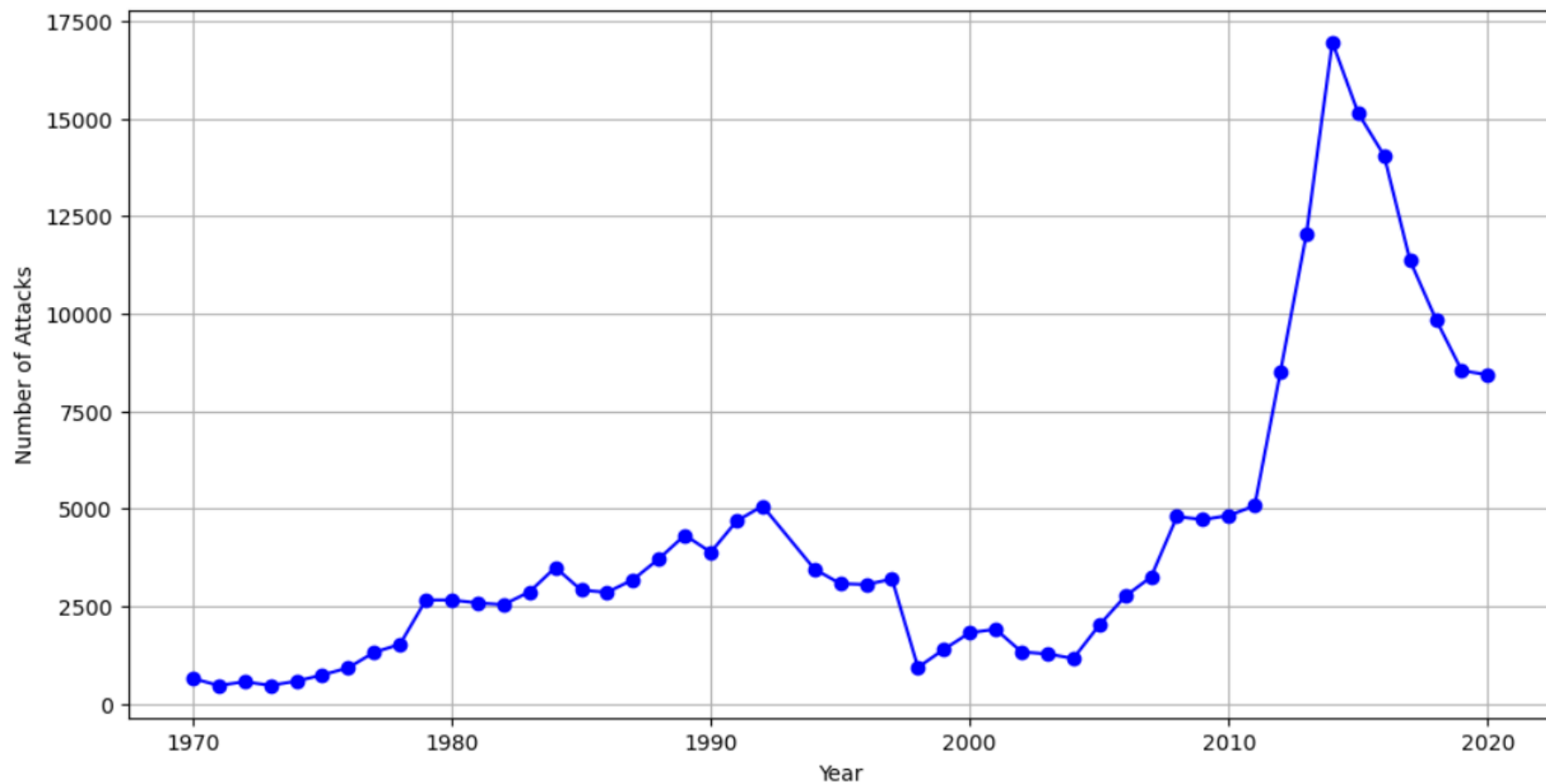
```
In [71]: 1 attacks_by_year
```

```
In [72]: 1 plt.figure(figsize=(12, 6))
2 attacks_by_year.plot(kind='line', marker='o', color='b')
3 plt.title('Number of Terrorist Attacks Per Year')
4 plt.xlabel('Year')
5 plt.ylabel('Number of Attacks')
6 plt.grid(True)
7 plt.show()
```

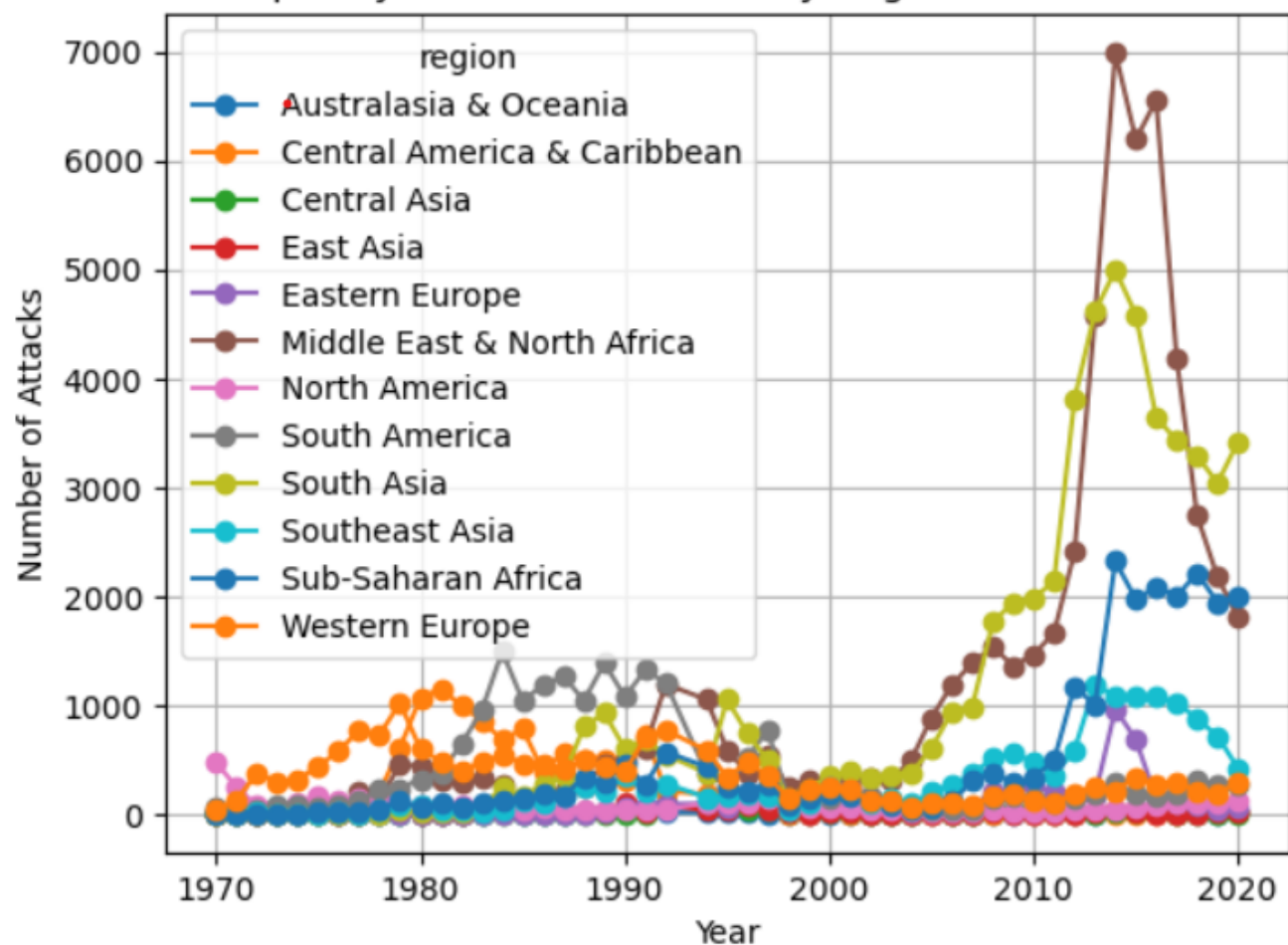
```
In [150]: 1 attacks_by_region = df.groupby(['year', 'region']).size().unstack(fill_value=0)
2 attacks_by_region
```

```
In [85]: 1 plt.figure(figsize=(12, 6))
2 attacks_by_region.plot(kind='line', marker='o')
3 plt.title('Frequency of Terrorist Attacks by Region Over the Years')
4 plt.xlabel('Year')
5 plt.ylabel('Number of Attacks')
6 plt.grid(True)
7 plt.show()
```

Number of Terrorist Attacks Per Year



Frequency of Terrorist Attacks by Region Over the Years

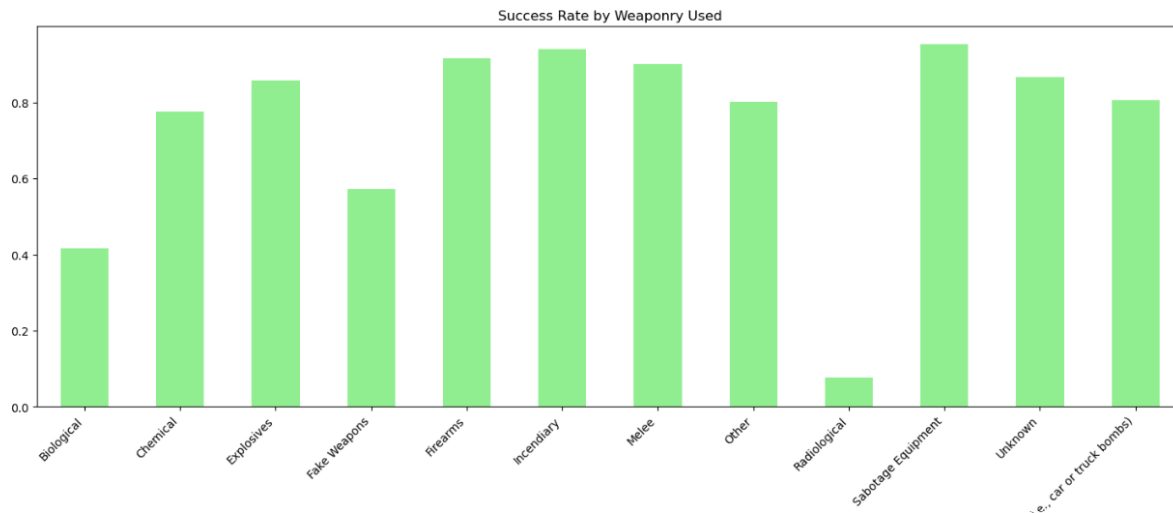
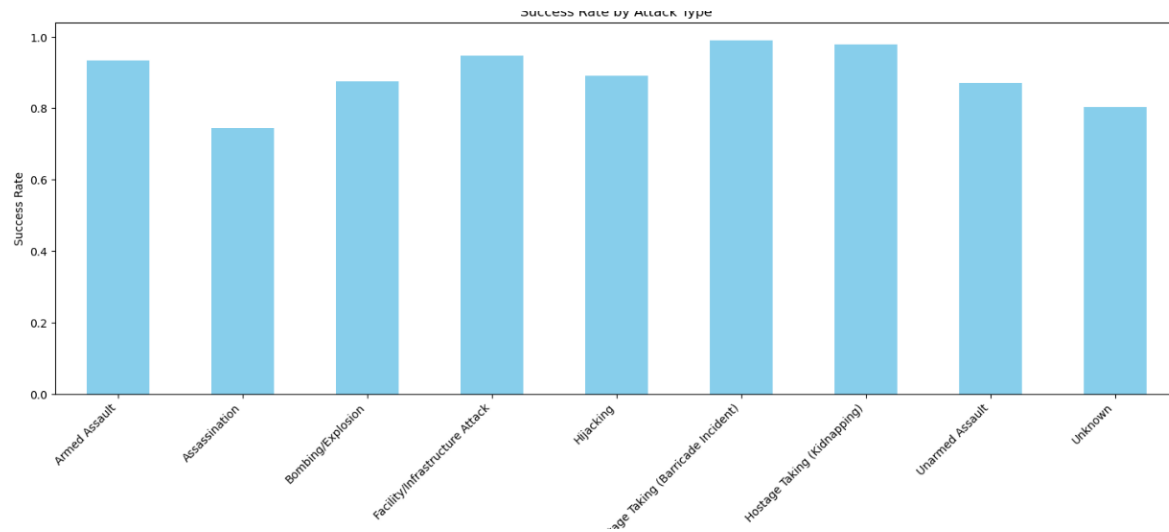


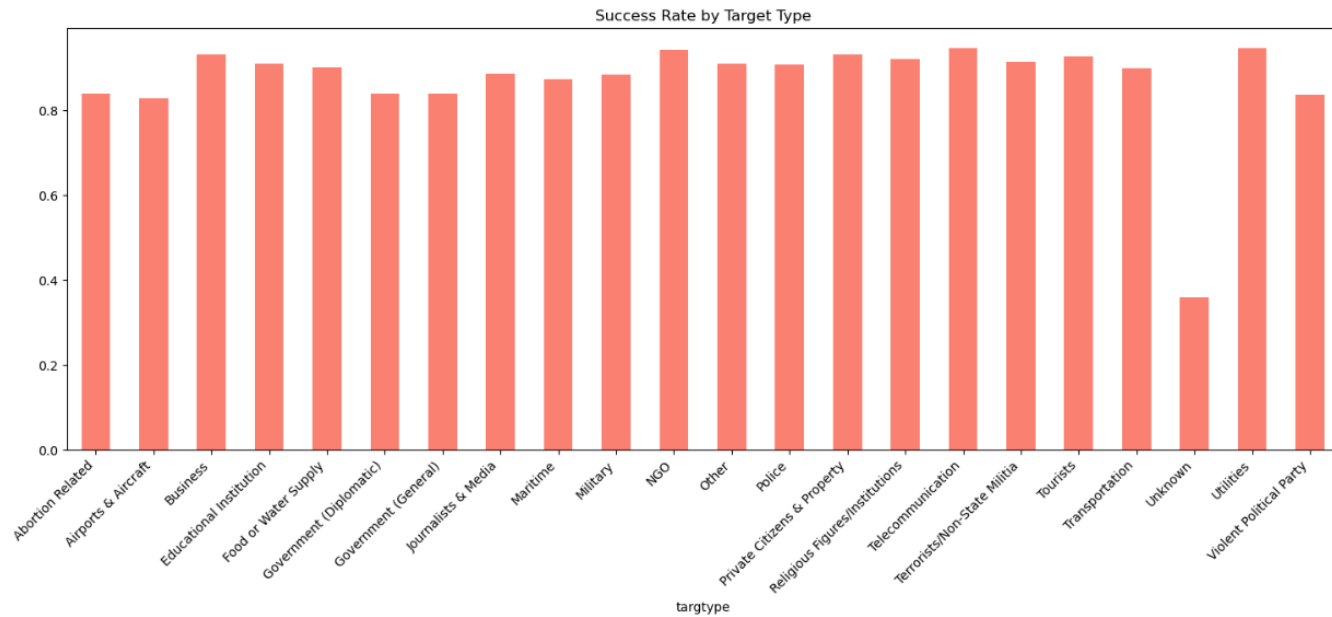
Effectiveness of Attacks: Analyse the success rate of terrorist attacks. What factors contribute to the success or failure of these attacks? Consider aspects such as the type of attack, weaponry used, and target type.

```
In [92]: 1 success_by_attack_type = df.groupby('attacktype')['success'].mean()
        2 success_by_attack_type
```

```
Out[92]: attacktype
Armed Assault                0.934757
Assassination                0.744324
Bombing/Explosion            0.874589
Facility/Infrastructure Attack 0.947343
Hijacking                    0.890646
Hostage Taking (Barricade Incident) 0.990484
Hostage Taking (Kidnapping)  0.978426
Unarmed Assault              0.871440
Unknown                      0.803509
Name: success, dtype: float64
```

```
In [98]: 1 plt.figure(figsize=(18, 6))
        2 success_by_attack_type.plot(kind='bar', color='skyblue')
        3 plt.title('Success Rate by Attack Type')
        4 plt.ylabel('Success Rate')
        5 plt.xticks(rotation=45, ha='right')
        6 plt.show()
```





Hostage taking and sabotage equipment are two potent strategies employed in attacks worldwide, leveraging the inherent risks and strategic deployment to achieve success. These tactics have demonstrated their effectiveness in various scenarios, showcasing their impact and versatility. The calculated use of such tools plays a crucial role in the success of these attacks, highlighting the importance of planning and execution. Hostage taking and sabotage are not only effective in causing harm but also in instilling fear and chaos, amplifying their power as tools of aggression. Their ability to disrupt and create turmoil makes them formidable weapons for individuals or groups aiming to achieve their objectives through forceful means.

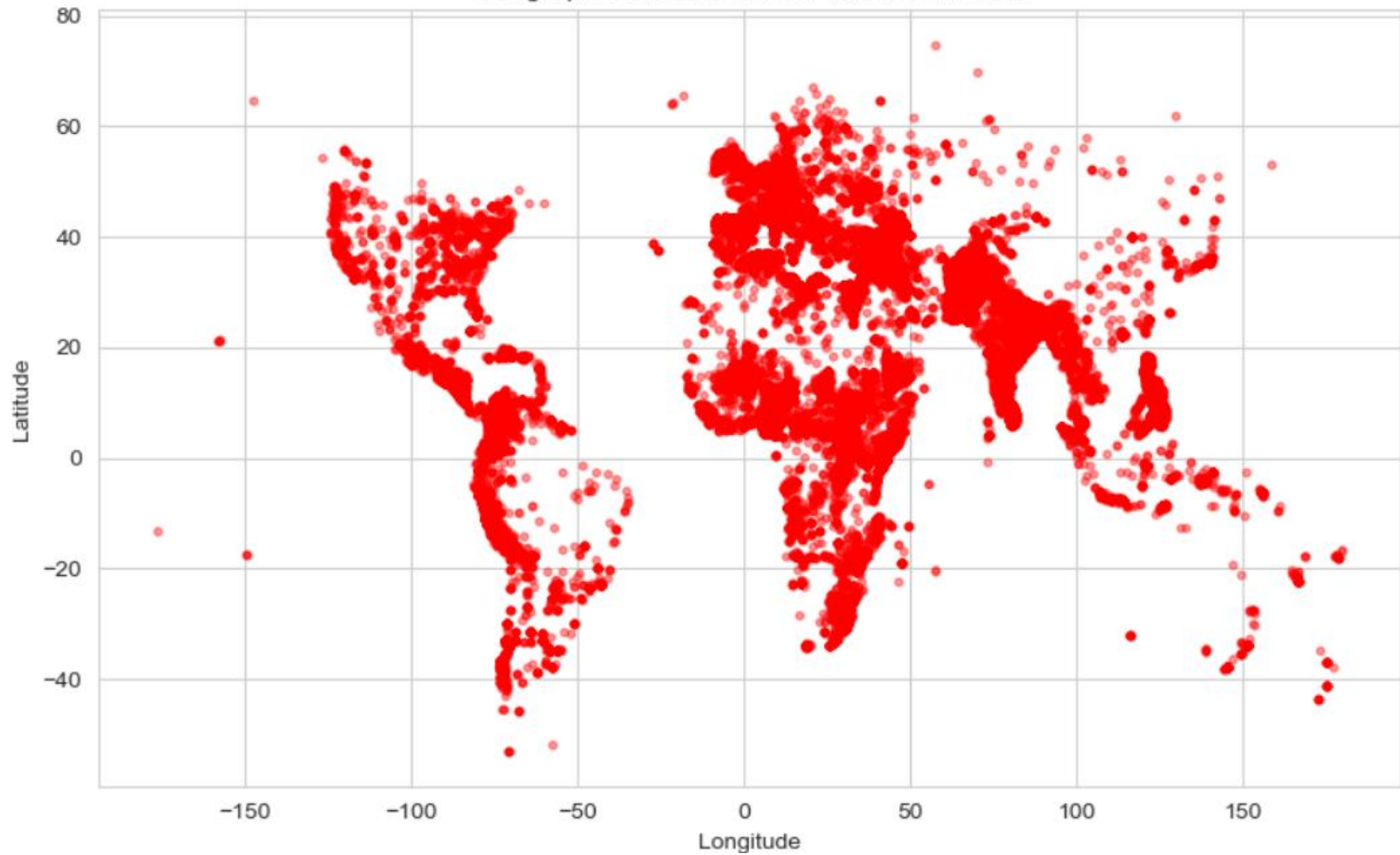
Geospatial Analysis: What are the geographical hotspots for terrorist activities?

Examine the concentration of attacks by latitude and longitude and identify if certain cities or provinces have higher incidences

```
In [162]: 1 import matplotlib.pyplot as plt
          2
          3 # Plotting all attacks based on latitude and longitude
          4 plt.figure(figsize=(10, 6))
          5 plt.scatter(df['longitude'], df['latitude'], alpha=0.4, color='red', s=10)
          6 plt.title('Geographical Distribution of Terrorist Attacks')
          7 plt.xlabel('Longitude')
          8 plt.ylabel('Latitude')
          9 plt.grid(True)
         10 plt.show()
         11
```

Geographically, the Middle East and North Africa region emerges as a significant hotspot for terrorist activities, characterized by a concentration of attacks in various cities and provinces. The latitude and longitude data reveal a clustering of incidents in this area, indicating a higher incidence of terrorism-related events. Cities like Baghdad, Kabul, and Damascus have experienced a notable frequency of attacks, underscoring their status as hotspots for terrorist activities. Provinces such as Anbar in Iraq and Kandahar in Afghanistan also stand out for their elevated levels of terrorist incidents. The complex socio-political landscape of the Middle East and North Africa contributes to the prevalence of terrorism in these regions, necessitating comprehensive strategies for addressing and mitigating such threats.

Geographical Distribution of Terrorist Attacks



Attack Types and Impact: Compare the different types of attacks (e.g., bombings, assassinations, kidnappings) and their outcomes in terms of fatalities and injuries.

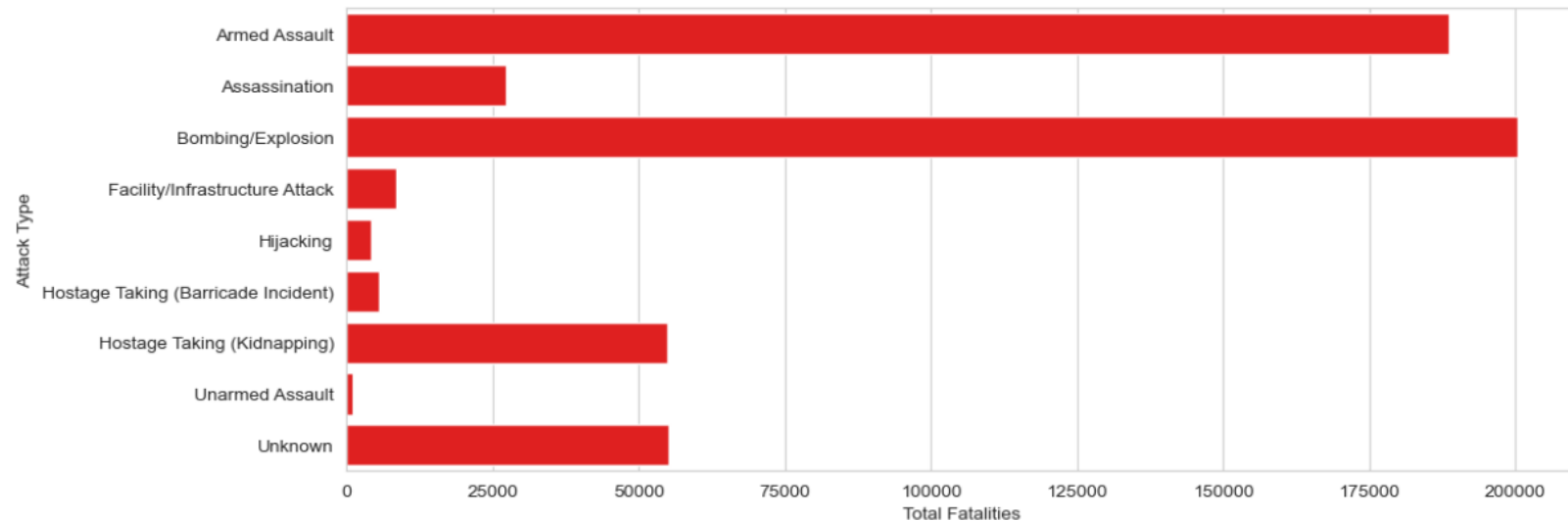
Which attack types are the deadliest?

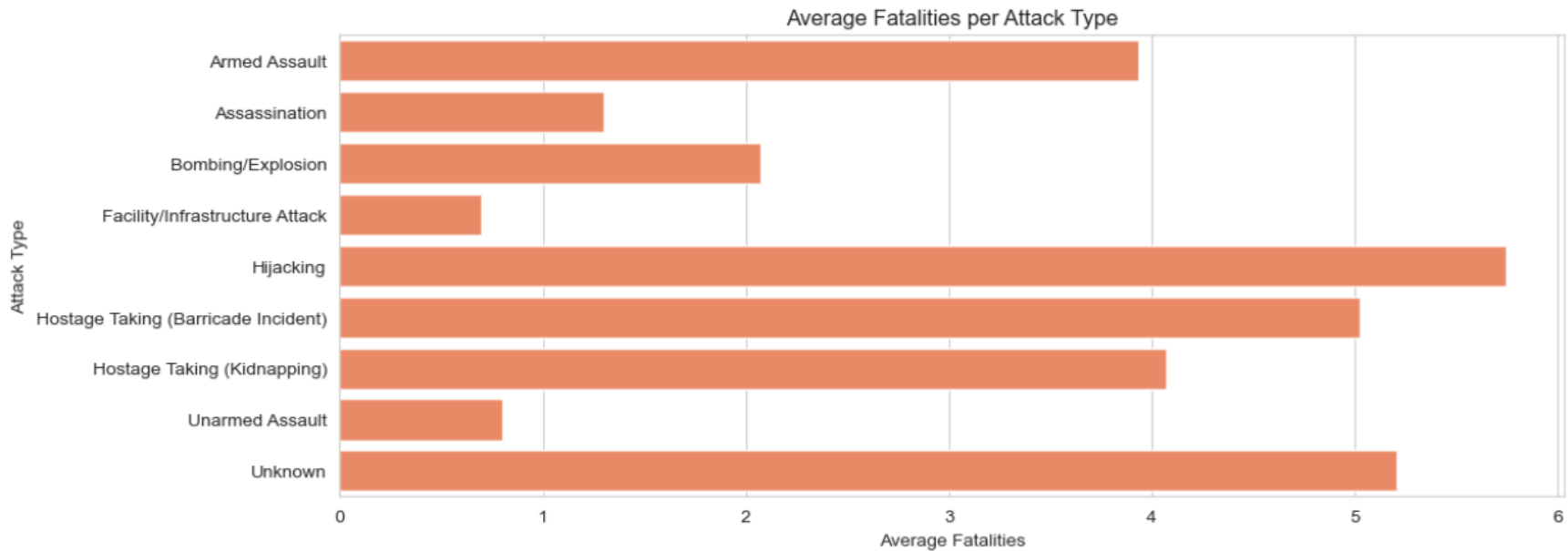
```
In [164]: 1 attack_stats = df.groupby('attacktype').agg({'kills': ['sum', 'mean']}).reset_index()
          2
          3 # Renaming columns for clarity
          4 attack_stats.columns = ['Attack Type', 'Total Fatalities', 'Average Fatalities']
          5
          6 # Display the statistics
          7 print(attack_stats)
```

	Attack Type	Total Fatalities	Average Fatalities
0	Armed Assault	188671.0	3.932448
1	Assassination	27375.0	1.301093
2	Bombing/Explosion	200461.0	2.073041
3	Facility/Infrastructure Attack	8491.0	0.698675
4	Hijacking	4183.0	5.745879
5	Hostage Taking (Barricade Incident)	5663.0	5.024845
6	Hostage Taking (Kidnapping)	54994.0	4.068506
7	Unarmed Assault	959.0	0.799833
8	Unknown	55077.0	5.209212

In [165]:

```
1 sns.set_style("whitegrid")
2
3 f, (ax1,ax2) = plt.subplots(2,1, figsize=(12,10),sharey=True)
4
5 g1=sns.barplot(x='Total Fatalities', y='Attack Type', data=attack_stats, color='red',ax=ax1)
6 plt.title('Total Fatalities by Attack Type')
7
8 g2=sns.barplot(x='Average Fatalities', y='Attack Type', data=attack_stats, color='coral',ax=ax2)
9 plt.title('Average Fatalities per Attack Type')
10
11 plt.show()
```





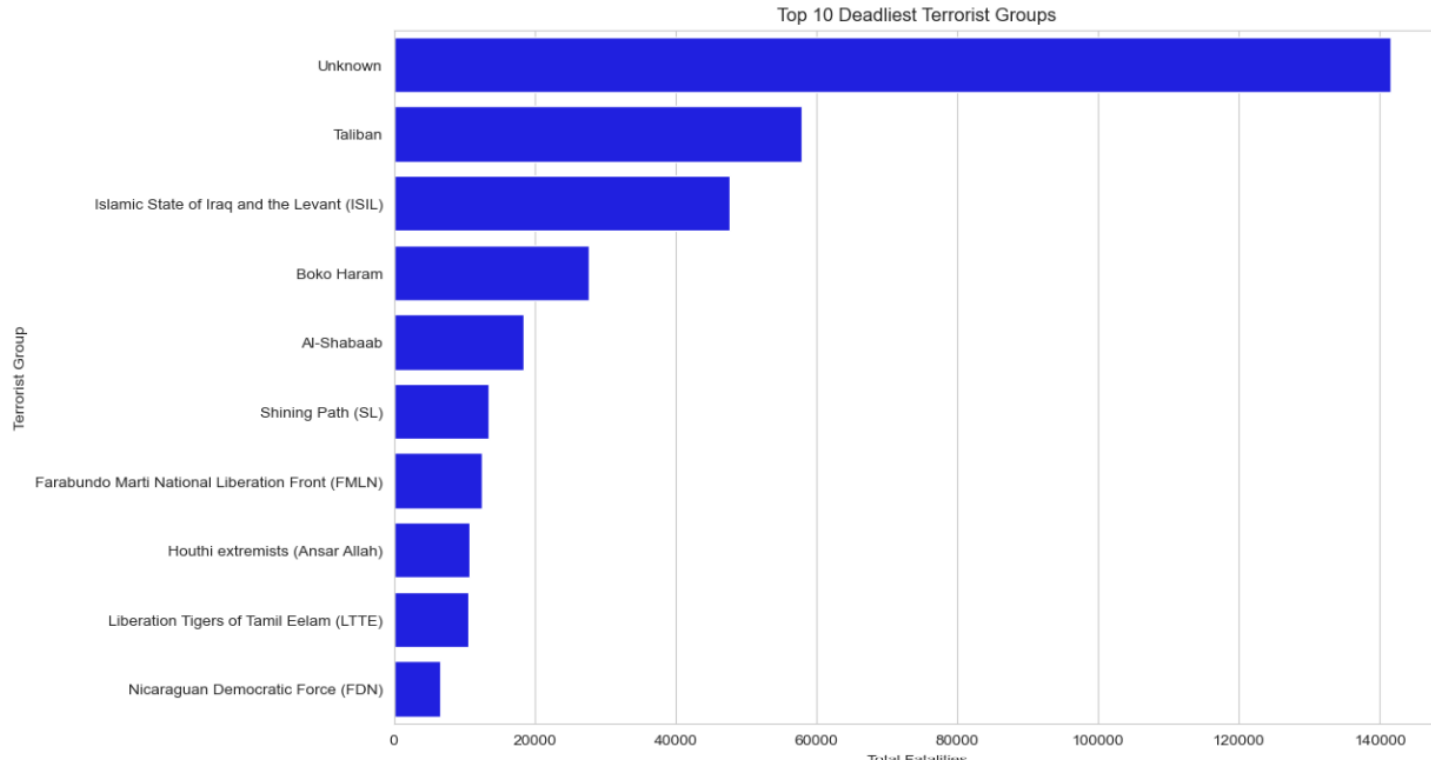
When comparing different types of attacks, armed assault and bombings emerge as the deadliest in terms of fatalities, often resulting in a significant loss of life due to the use of weapons and explosives. These types of attacks tend to cause widespread devastation and casualties. On the other hand, hijacking and hostage situations, while less frequent, have a higher average number of fatalities per incident, as they involve the direct targeting and endangerment of individuals. The nature of these attacks, which often involve prolonged standoffs and negotiations, can lead to tragic outcomes for those involved. Understanding the distinct characteristics and impacts of each attack type is crucial for developing effective counterterrorism strategies and enhancing security measures to prevent such incidents.

Group Analysis: Which terrorist groups are the most active? Analyze the data to see which groups carry out the most attacks and which groups result in the highest number of fatalities.

```
In [168]: 1 group_stats = df.groupby('gang').agg({'gang': 'count', 'kills': 'sum'}).rename(columns={'gang': 'Number of Attacks', 'kills':  
2 group_stats.sort_values(by='Number of Attacks', ascending=False, inplace=True)  
3 group_stats
```

Out[168]:

	gang	Number of Attacks	Total Fatalities
3515	Unknown	90580	141603.0
3245	Taliban	11954	57931.0
1580	Islamic State of Iraq and the Levant (ISIL)	7220	47696.0
182	Al-Shabaab	4403	18450.0
3048	Shining Path (SL)	4315	13466.0
...	...	...	...
821	Christian Palestinian group	1	0.0
822	Christian State-Holy Rus	1	0.0
2185	Muslim Commando (Iraqi revolutionary brothers)	1	4.0
825	Chukakuha Revolutionary Army	1	0.0
3651	leftist guerrillas-Bolivarian militia	1	0.0



Based on the data analysis, the Taliban emerges as the most active and deadliest terrorist group. The Taliban has been responsible for a significant number of attacks, carrying out operations with high frequency and intensity. Their tactics often result in a high number of fatalities, making them a major threat to security and stability in the regions where they operate

Weapon Types: What are the most common weapon types used in terrorist attacks?

Investigate how the choice of weapon correlates with the number of kills and the success rate of attacks

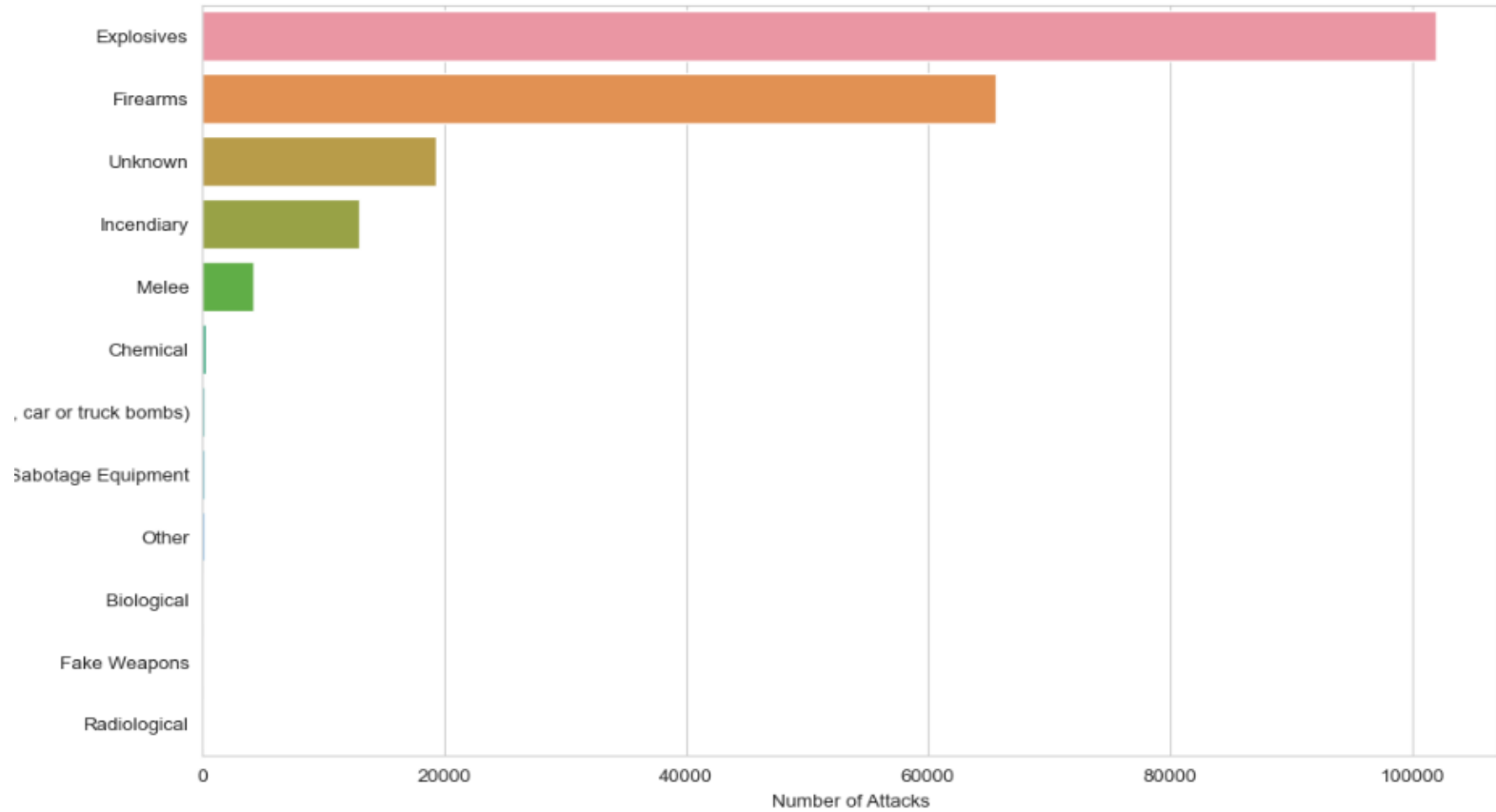
```
In [170]: 1 weapon_stats = df.groupby('weaptype').agg({'eventid': 'count', 'kills': 'mean', 'success': 'mean'}).rename(columns={'eventid': 'Number of Attacks'})
2
3 # Sort the results by the number of attacks
4 weapon_stats.sort_values(by='Number of Attacks', ascending=False, inplace=True)
```

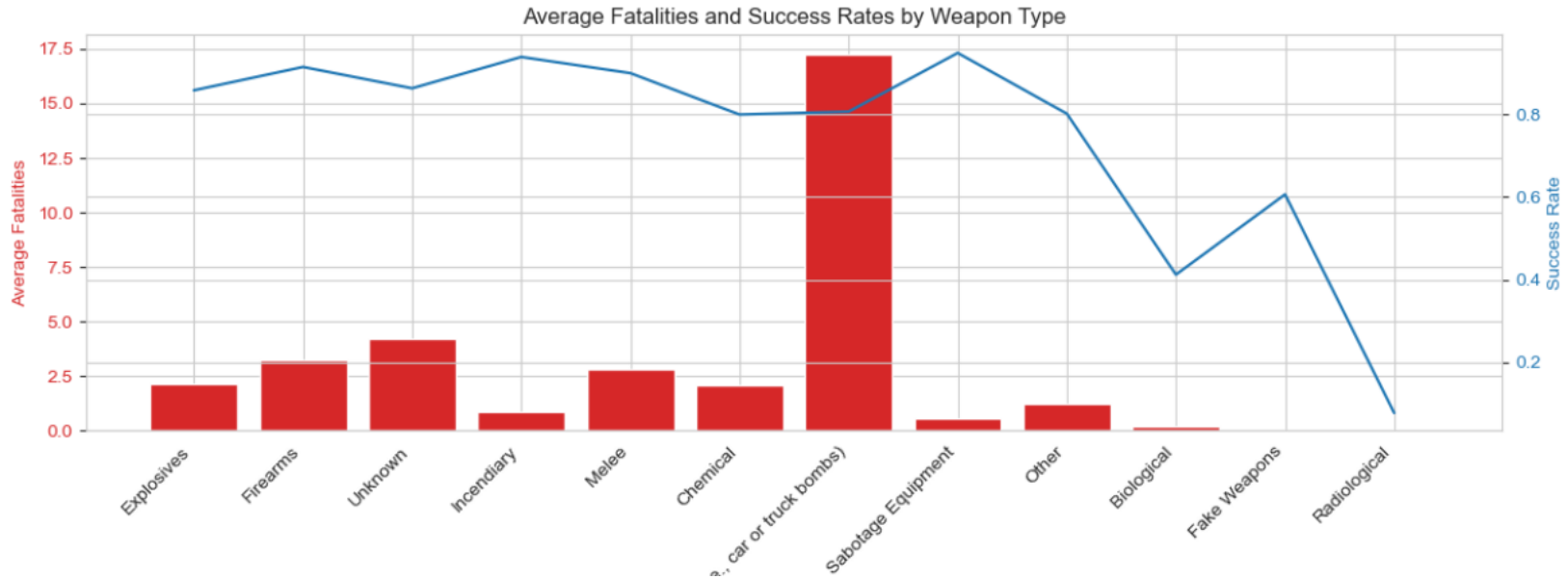
```
In [171]: 1 weapon_stats
```

Out[171]:

	weaptype	Number of Attacks	Average Fatalities	Success Rate
2	Explosives	102008	2.185878	0.858109
4	Firearms	65653	3.267924	0.914657
10	Unknown	19300	4.199119	0.863005
5	Incendiary	12938	0.863503	0.938940
6	Melee	4219	2.839535	0.899739
1	Chemical	319	2.081505	0.799373
11	Vehicle (not to include vehicle-borne explosiv...	186	17.274194	0.806452
9	Sabotage Equipment	175	0.560000	0.948571
7	Other	136	1.250000	0.801471
0	Biological	34	0.205882	0.411765
3	Fake Weapons	33	0.030303	0.606061
8	Radiological	13	0.000000	0.076923

Most Common Weapon Types Used in Terrorist Attacks





Explosives are indeed the most used weapon type in terrorist attacks, due to their destructive capabilities and relative ease of deployment. Among explosive devices, car or truck bombs stand out for their devastating impact, often resulting in the highest average number of fatalities in attacks where they are employed. The use of vehicles as weapons allows terrorists to cause mass casualties in crowded areas. On the other hand, sabotage equipment, while less common, has a high success rate in achieving the intended objectives of the attackers. This could be attributed to the stealthy nature of sabotage operations and the potential for causing significant damage without direct confrontation. Understanding the correlation between weapon types and their outcomes is crucial for developing effective counterterrorism strategies and enhancing security measures to mitigate the impact of terrorist attacks.

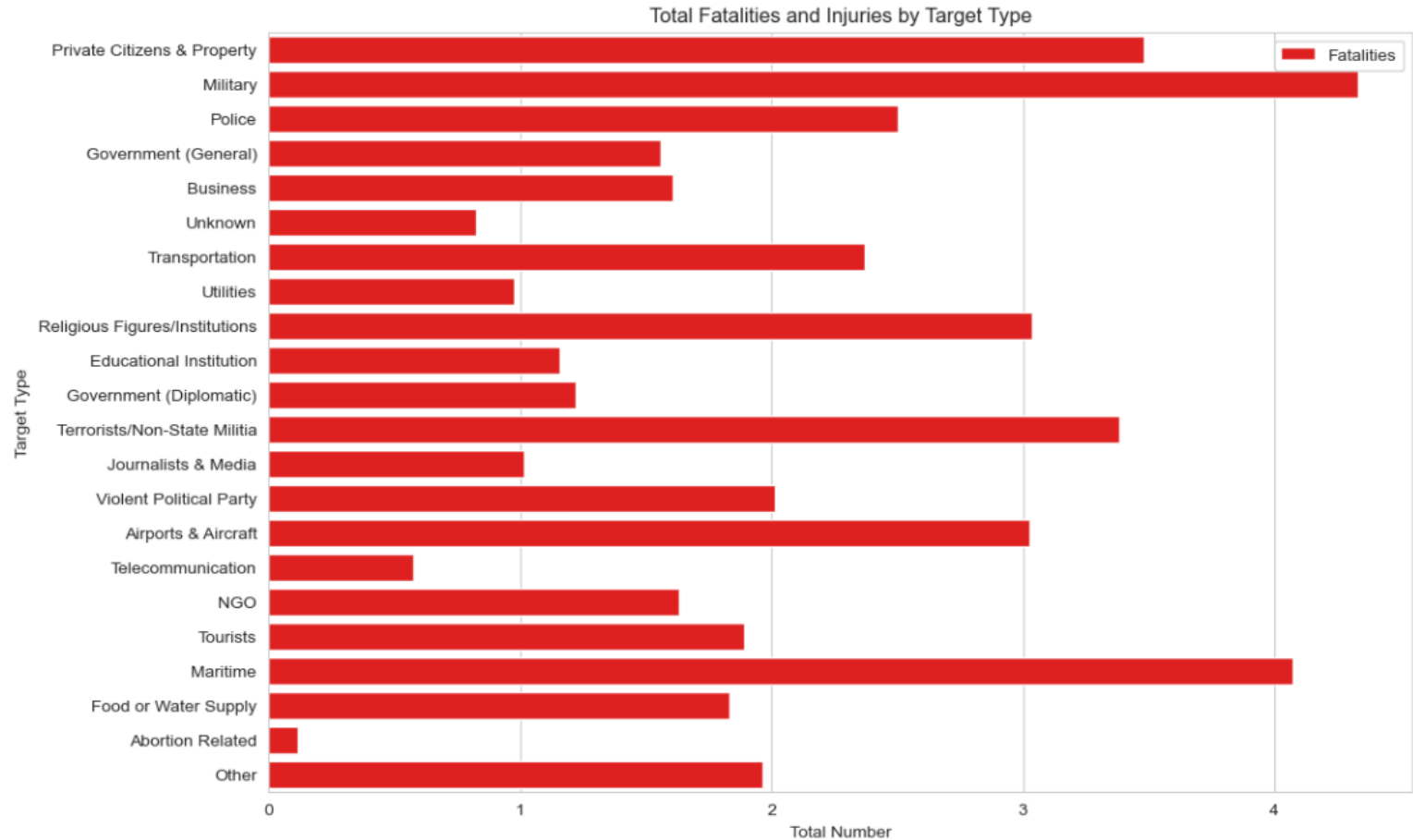
Target Analysis: What types of targets are most frequently attacked? Assess the vulnerability of different target types like government facilities, private citizens, or infrastructure.

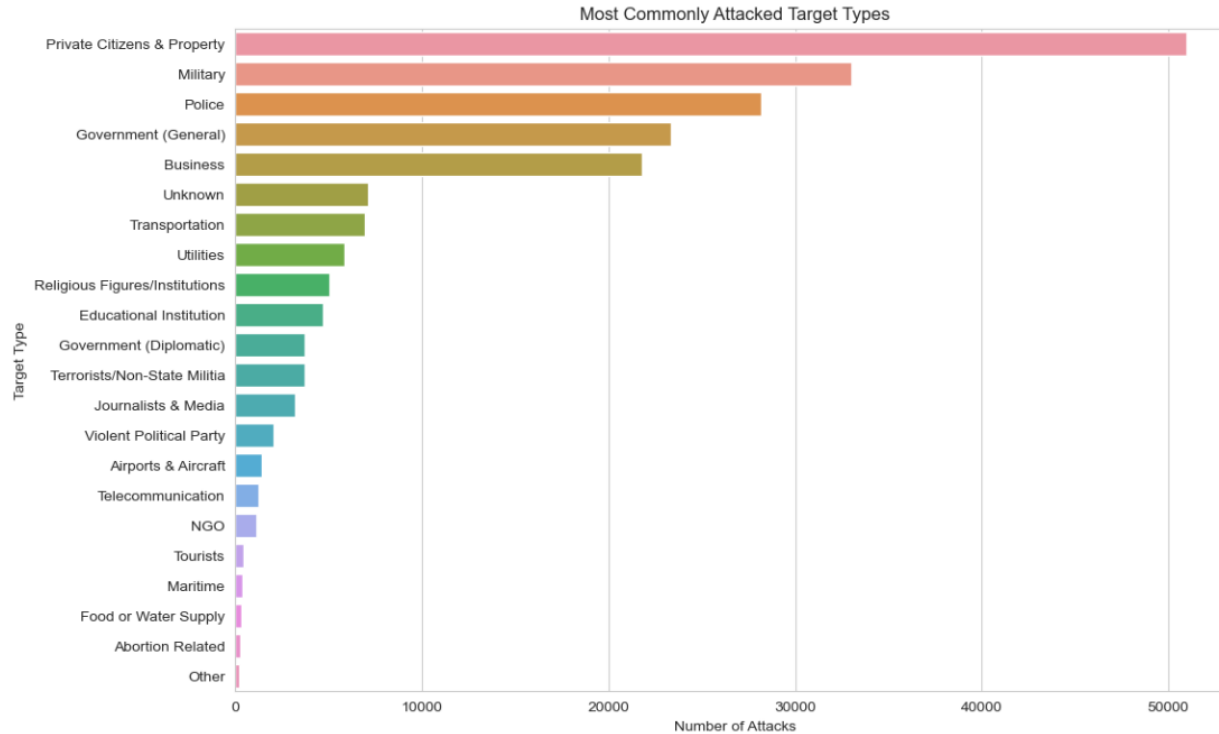
```
In [148]: 1 # Aggregating data by target type
2 target_stats = df.groupby('targettype').agg({'targettype': 'count', 'kills': 'mean'}).rename(columns={'targettype': 'Number of At
3
4 # Sort the results by the number of attacks
5 target_stats.sort_values(by='Number of Attacks', ascending=False, inplace=True)
6
```

```
In [151]: 1 target_stats
```

```
In [156]: 1 import matplotlib.pyplot as plt
2 import seaborn as sns
3
4 # Setting the aesthetic style of the plots
5 sns.set_style("whitegrid")
6
7 # Plotting the number of attacks by target type
8 plt.figure(figsize=(12, 8))
9 sns.barplot(x='Number of Attacks', y='targettype', data=target_stats)
10 plt.title('Most Commonly Attacked Target Types')
11 plt.xlabel('Number of Attacks')
12 plt.ylabel('Target Type')
13 plt.show()
14
15 # Plotting total fatalities and injuries by target type
16 plt.figure(figsize=(12, 8))
17 sns.barplot(x='Average Fatalities', y='targettype', data=target_stats, color='red', label='Fatalities')
18 plt.title('Total Fatalities and Injuries by Target Type')
19 plt.xlabel('Total Number')
20 plt.ylabel('Target Type')
21 plt.legend()
22 plt.show()
23
```

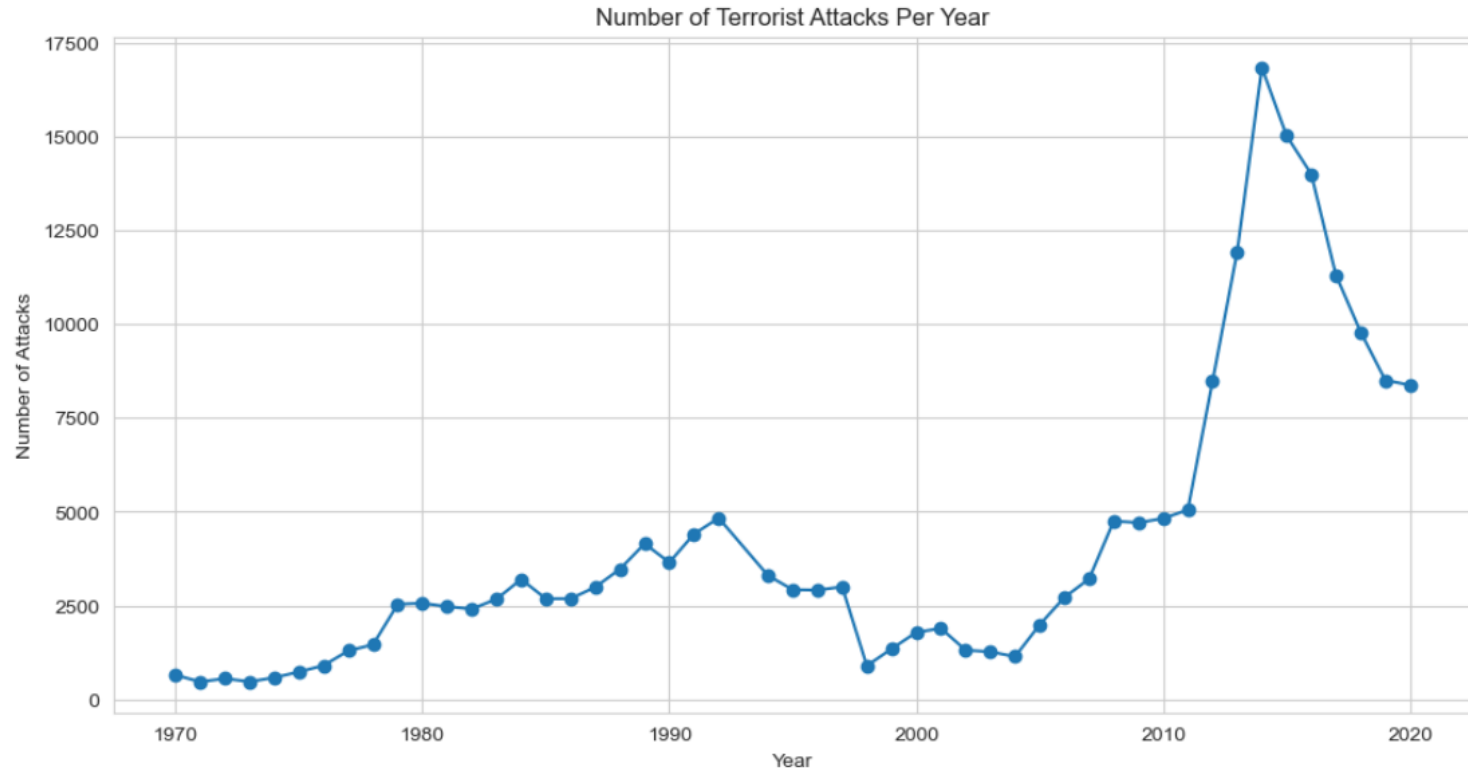
	targettype	Number of Attacks	Average Fatalities
13	Private Citizens & Property	50984	3.481112
9	Military	33013	4.333929
12	Police	28199	2.501153
6	Government (General)	23346	1.559539
2	Business	21777	1.608073
19	Unknown	7090	0.825529
18	Transportation	6919	2.372164
20	Utilities	5854	0.975060
14	Religious Figures/Institutions	5046	3.037852
3	Educational Institution	4706	1.158734
5	Government (Diplomatic)	3741	1.221331
16	Terrorists/Non-State Militia	3706	3.385861
7	Journalists & Media	3224	1.014578
21	Violent Political Party	2052	2.013645
1	Airports & Aircraft	1393	3.028715
15	Telecommunication	1262	0.575277
10	NGO	1098	1.630237
17	Tourists	442	1.893665
8	Maritime	363	4.074380
4	Food or Water Supply	335	1.832836
0	Abortion Related	267	0.116105
11	Other	197	1.964467

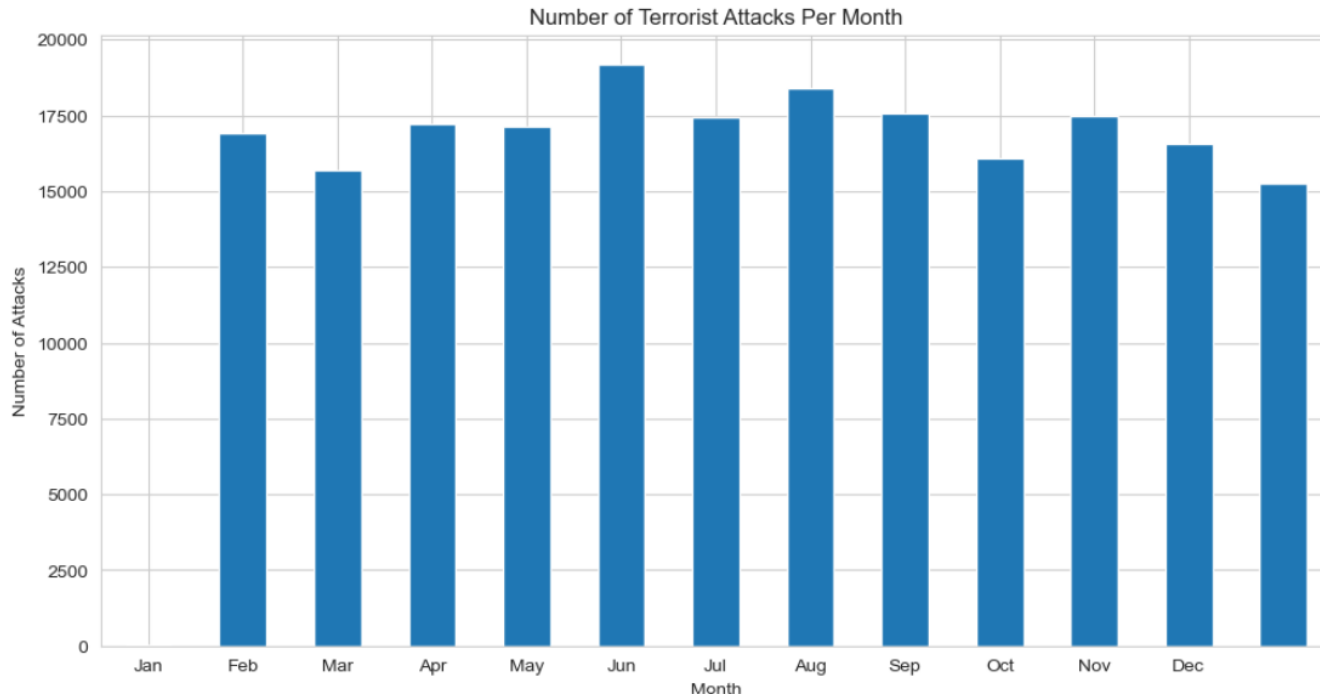




Based on the data analysis, it is evident that military targets experience the highest number of fatalities in terrorist attacks. This is often due to the strategic importance of military installations and the potential for large-scale casualties in such attacks. On the other hand, private civilians and property are the most frequently targeted by terrorists, reflecting the desire to instill fear and disrupt societal norms.

Temporal Patterns: Are there specific times of the year, month, or week when terrorist attacks are more frequent? Analyse the timing of attacks to identify any patterns.





The analysis of temporal patterns in terrorist attacks reveals that 2014 stands out as the year with the highest frequency of attacks. This peak could be attributed to specific geopolitical events or escalations in conflicts within certain regions during that year. Such a spike often indicates significant political or social unrest and may have involved the emergence or resurgence of terrorist groups. Further, the month of June is identified as having the highest number of attacks compared to other months. This finding could suggest a strategic choice, potentially related to weather conditions that are favourable for mobility and operations. June also marks the midpoint of the year, possibly aligning with tactical reassessments and campaign intensifications by terrorist organizations.

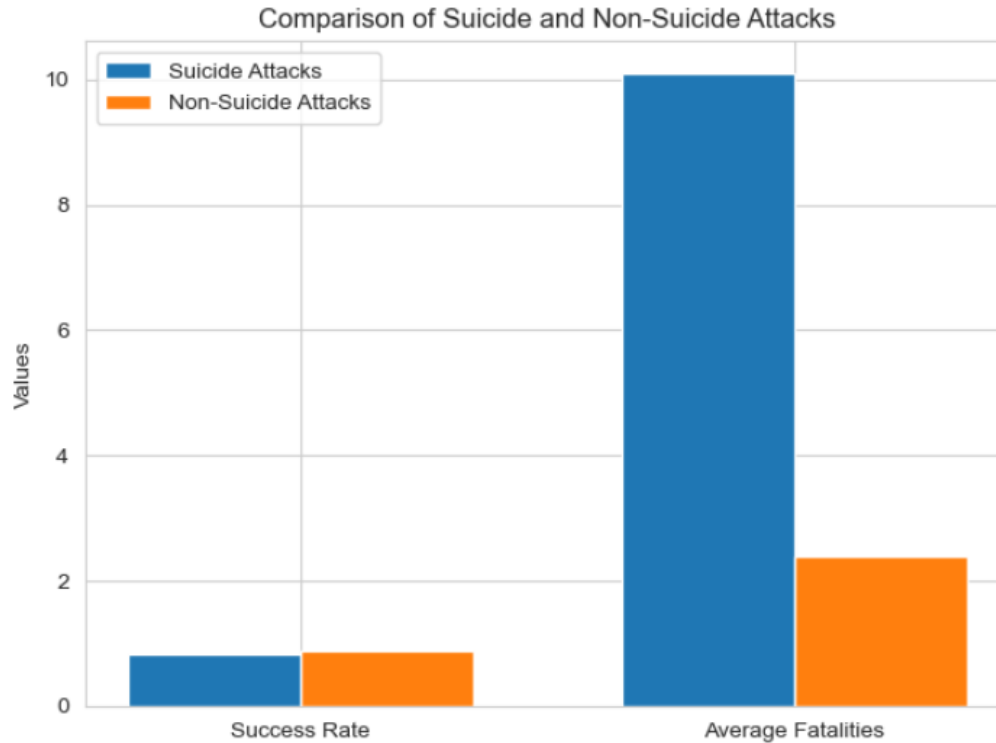
Suicide Attacks: How prevalent are suicide attacks, and how do their outcomes compare to no suicide attacks in terms of success rate and casualties?

```
In [175]: 1 suicide_attacks = df[df['suicide'] == 1]
          2 non_suicide_attacks = df[df['suicide'] == 0]
```

```
In [176]: 1 # Calculate success rates, average fatalities, and injuries
          2 suicide_stats = {
          3     'Success Rate': suicide_attacks['success'].mean(),
          4     'Average Fatalities': suicide_attacks['kills'].mean(),
          5 }
          6
          7 non_suicide_stats = {
          8     'Success Rate': non_suicide_attacks['success'].mean(),
          9     'Average Fatalities': non_suicide_attacks['kills'].mean(),
         10 }
         11
         12 print("Suicide Attack Stats:", suicide_stats)
         13 print("Non-Suicide Attack Stats:", non_suicide_stats)
         14
```

Suicide Attack Stats: {'Success Rate': 0.8362557323981656, 'Average Fatalities': 10.104127326679256}

Non-Suicide Attack Stats: {'Success Rate': 0.8841042510121457, 'Average Fatalities': 2.383410931174089}



The fatalities in suicide attacks are typically higher than in non-suicide attacks due to the intentional targeting of crowded areas. However, the success rate of suicide attacks is comparable to non-suicide attacks, with both types achieving their intended objectives in a similar proportion of cases. Non-suicide attacks may have a slight edge in terms of success rate, possibly due to the element of surprise or the ability to escape after the attack.

03.

Conclusion

In conclusion, the analysis of weapon types and target vulnerabilities in terrorist attacks provides valuable insights into the nature of these threats. The prevalence of explosives, especially car or truck bombs, underscores the destructive potential of such weapons and the urgent need for robust security measures to prevent their use by terrorists. Military targets emerge as high-risk locations, with the highest number of fatalities recorded in attacks on these installations, highlighting the strategic importance of safeguarding military facilities. Conversely, private civilians and property are frequently targeted by terrorists, indicating a deliberate intent to instill fear and disrupt societal stability. The vulnerability of these targets underscores the importance of enhancing security measures to protect vulnerable populations and critical infrastructure. Understanding the patterns and trends in terrorist attacks is essential for developing effective counterterrorism strategies that address the diverse threats posed by different weapon types and target types. By prioritizing resources and security measures based on these insights, authorities can better prepare and respond to terrorist threats, ultimately enhancing the safety and security of communities and mitigating the devastating impact of terrorism on society.

Thankyou